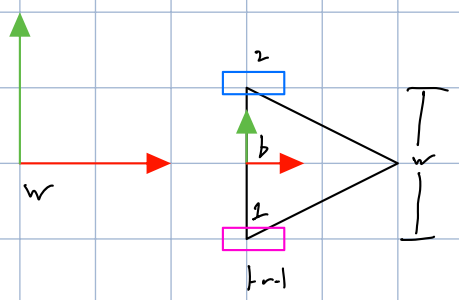


$$T_{wb} = \begin{matrix} \theta \\ x \\ y \end{matrix}$$

$$T_{b_1} = \begin{matrix} 0 \\ -w/2 \\ 0 \end{matrix} \qquad Ad_{b1} \begin{matrix} 1 & 0 & 0 \\ -w/2 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

$$T_{b_2} = \begin{matrix} 0 \\ w/2 \\ 0 \end{matrix} \qquad Ad_{b2} \begin{matrix} 1 & 0 & 0 \\ w/2 & 1 & 0 \\ 0 & 0 & 1 \end{matrix}$$

$$V_x = A_{xb} V_b = \begin{matrix} 0 + \dot\theta \\ \mp\dot\theta \cdot \frac{w}{2} + V_x \\ 0 + V_y \end{matrix}$$

$$\mp\dot\theta\frac{w}{2} + V_x = r\dot\phi_{1,2}, \quad V_y = 0$$

$$\dot\phi_{1,2} = \begin{matrix} -\frac{w}{\pm 2r} & \frac{1}{r} & 0 \\ 0 & 0 & 1 \end{matrix} \begin{bmatrix} \theta \\ V_x \\ V_y \end{bmatrix}$$

$$u \qquad H \qquad V_b$$

$$\begin{matrix} \phi_1 \\ \phi_2 \end{matrix} = \frac{1}{r} \begin{bmatrix} -\frac{w}{2} & 1 & 0 \\ \frac{w}{2} & 1 & 0 \end{bmatrix} \begin{matrix} \dot\theta \\ V_x \\ V_y \end{matrix} \Rightarrow \frac{r}{2} \begin{bmatrix} -\frac{1}{w} & \frac{1}{w} \\ 1/2 & 1/2 \\ 0 & 0 \end{bmatrix} \begin{matrix} \dot\phi_1 \\ \dot\phi_2 \end{matrix} \Rightarrow$$

$$V_b = H^+ u$$

$$V_w = A_{wb} \overset{H^+ u}{V_b}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -y\cos\theta + x\sin\theta & \cos\theta & \sin\theta \\ x\cos\theta + y\sin\theta & -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} -\frac{1}{w} & \frac{1}{w} \\ 1/2 & 1/2 \\ 0 & 0 \end{bmatrix} \begin{matrix} \dot\phi_1 \\ \dot\phi_2 \end{matrix}$$

$$= w(\phi_2 - \phi_1)$$

$$\phi_1(-wxs + wyc + 2c) + \phi_2(wxs - wyc + 2c)$$

$$\phi_1(-wxc - wys - 2s) + \phi_2(wxc + wys - 2s)$$

$$\begin{matrix} 0 \\ 0 \end{matrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \theta \\ V_x \\ V_y \end{matrix} \Rightarrow V_y \text{ must } = 0$$

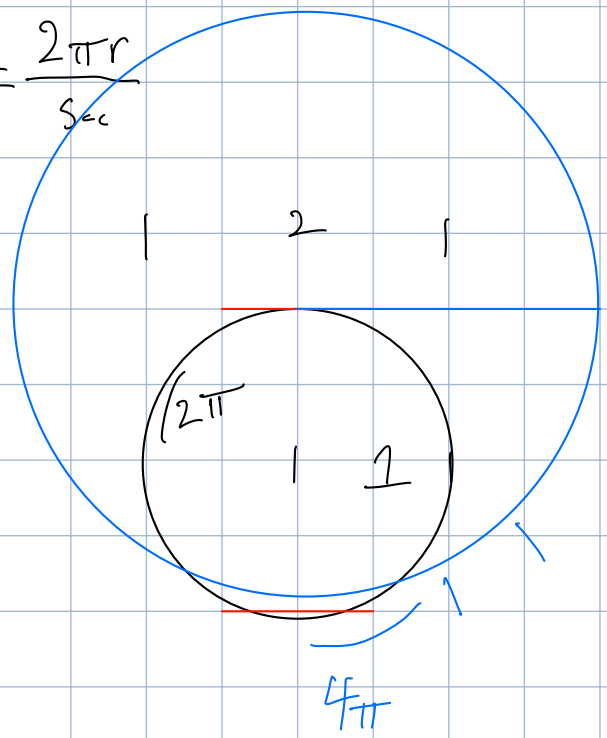
$$\frac{r}{2} \frac{-(\phi_1 - \phi_2)}{w}$$

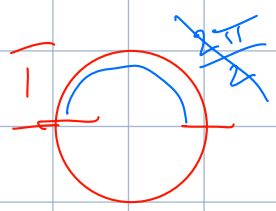
$$\frac{\rho_1 + \rho_2}{2}$$

$$\frac{1}{r}\left(x - \theta\frac{w}{2}\right) \quad \frac{1}{r}\left(x + \theta\frac{w}{2}\right)$$

right left

$$\frac{2\pi \text{ rad}}{1 \text{ sec}} \cdot \frac{\cancel{2\pi} \cdot r \text{ m}}{\cancel{2\pi} \text{ rad}} = \frac{2\pi r}{\text{sec}}$$





wheel rad = 1

$\frac{\pi}{2}$ rotation $\left(\frac{1}{4} \text{ rot}\right) = \frac{2\pi}{4} = \frac{1}{2}\pi$ length

path rad = 2

circ = $4\pi = \frac{1}{2}\pi \cdot 8$